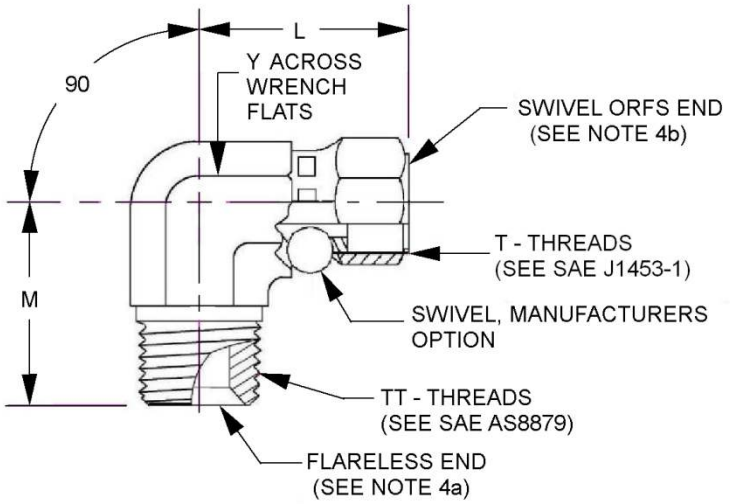
DETAIL SPECIFICATION SHEET

FITTING, ADAPTER, 90° ELBOW, HYDRAULIC AND PNEUMATIC, FLARELESS, MALE TO ORFS, FEMALE SWIVEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32480.



Dash size	Tube OD nom. inches (mm)	T ORFS SAE J1453-1	TT Flareless SAE AS8879
-4	.2500 (6.350)	0.5625-18UNF-2A	.4375-20 UNJF-3A
-6	.3750 (9.525)	0.6875-16UN-2A	.5000-20 UNJF-3A
-8	.5000 (12.700)	0.8125-16UN-2A	.7500-16 UNJF-3A
-10	.6250 (15.875)	1.0000-14UN-2A	.8750-14 UNJF-3A
-12	.7500 (19.050)	1.1875-12UN-2A	1.0625-12 UNJ-3A
-16	1.0000 (25.400)	1.4375-12UN-2A	1.3125-12 UNJ-3A
-20	1.2500 (31.750)	1.6875-12UN-2A	1.6250-12 UNJ-3A
-24	1.5000 (38.100)	2.0000-12UN-2A	1.8750-12 UNJ-3A

FIGURE 1. 90° elbow, flareless to ORFS adapter.

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Dash size	Tube OD nom. inches (mm)	Y – Hex Across flats +.0, -.03 (+.0, -0.8)	L inches (mm) ±.04 (1.0)	M inches (mm) ±.04 (1.0)
-4	.2500 (6.350)	.4375 (11.113)	.846 (21.50)	.890 (22.60)
-6	.3750 (9.525)	.5625 (14.285)	.984 (25.00)	1.060 (26.90)
-8	.5000 (12.700)	.7500 (19.050)	1.102 (28.00)	1.250 (31.80)
-10	.6250 (15.875)	.8750 (22.225)	1.319 (33.50)	1.450 (36.80)
-12	.7500 (19.050)	1.0625 (26.988)	1.476 (37.50)	1.660 (42.20)
-16	1.0000 (25.400)	1.3125 (33.338)	1.634 (41.50)	1.810 (46.00)
-20	1.2500 (31.750)	1.6250 (41.275)	1.752 (44.50)	2.060 (52.30)
-24	1.5000 (38.100)	1.8750 (47.625)	1.929 (49.00)	2.330 (59.20)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Break all sharp edges and remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with the following:
   a. SAE AS4375 style G for flareless end.
   b. SAE J1453-3*520221* for ORFS swivel end.
   c. SAE J514*070202* for body.

FIGURE 1. 90° elbow, flareless to ORFS adapter - Continued.

REQUIREMENTS:

Fittings shall be as specified on figure 1 and in table I.

Materials shall be in accordance with MIL-DTL-32480.

Material and finish designators. Material and finish designators shall be as specified in table I. All platings shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of corrosion after 96 hours of salt spray. Fluid passages, other openings, and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

Dimensions. All dimensions not specified on figure 1 shall be in accordance with SAE AS4375 for the flareless end, SAE J1453-3 for the ORFS end and SAE J514 for the body.

Across flat widths. Across flat widths shall fit standard wrench openings.
Across flat widths shall be in accordance with applicable style code SAE J514 90º elbow.

Inside diameter. The inside diameter shall be in accordance with SAE AS4375 for the flareless end and in SAE J1545-3 for the ORFS end, drill depth shall be optional with manufacturer. Wall thickness shall be in accordance with SAE AS4375 for the flareless end and SAE J1453-3 for the ORFS end.

Body thickness. The radius and body thickness shall conform to SAE J514.

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TABLE I. Material and chemical finish identification codes and chemical finish reference specifications. 1/

Material and finish code	Material	Plating finish
Blank	Steel	Cadmium plating in accordance with SAE AMS-C-81562, type II, class 3 or SAE AMS-QQ-P-416, type II, class 2. 2/
CN		Cadmium plating in accordance with SAE AMS-C-81562, type II, class 3 or SAE AMS-QQ-P-416, type II, class 2 with NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706, type II, class 1A.
E		NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A.
F	Steel	Zinc plate (finish J, P, or R) with NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A.
H	Steel	Aluminum-nickel in accordance with ASTM F1136/F1136M, grade 3, NC.
J	Steel	Zinc-nickel in accordance with SAE AMS2417, type 2, grade B.
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class1.
R	Steel	Zinc plating in accordance with ASTM B633; type VI, Fe/Zn 5. 3/
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE AMS2700, type 6 or 7.
T	Titanium 4/	Anodize in accordance with SAE AMS2488 type 2.
TF		Fluoride phosphate in accordance with SAE AMS2486.
Z	Steel	Zinc plating in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5.
ZN	Steel	Zinc plating in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5 with NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A.

1/ Materials and finishes shall be in accordance with MIL-DTL-32480.
2/ Hydrogen embrittlement relief test need not be run.
3/ Hexavalent chromium free.
4/ Titanium shall not be used in oxygen systems.

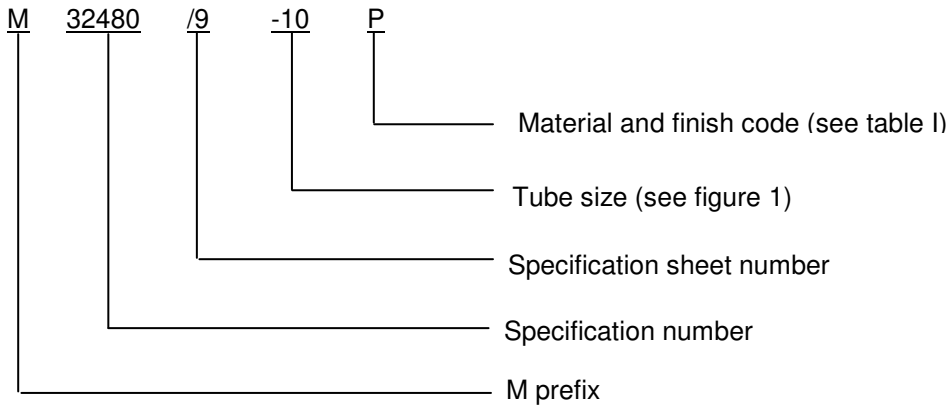
Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using the brush plating process below. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE AMS2451/5.
b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE AMS2451/9.
c. Brush plating of NAVAIR TCP shall be in accordance with MIL-DTL-81706, type II, class 1A, material form 1 through 6, application method B. Example of a PIN: M817062A6B.

Maximum operating pressure. Maximum operating pressure shall be in accordance with MIL-DTL-32480.

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Part or Identifying Number (PIN): The PIN consists of the letter M, the basic specification number, the specification sheet number, fitting dash size, and a letter for the material finish designator.



PIN example: M32480/9-10P indicates a 90°, male to female, adapter, .6250 inch (15.875 mm), steel with zinc phosphate finish.

O-ring is not supplied. Order separately, see MIL-DTL-25988.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Referenced documents. In addition to MIL-DTL-32480, this document references the following:

MIL-DTL-16232	ASTM F1136/F1136M	SAE AMS2488
MIL-DTL-81706	SAE AMS-C-81562	SAE AMS2700
MIL-DTL-25988	SAE AMS-QQ-P-416	SAE AS4375
ASTM B117	SAE AMS2417	SAE AS8879
ASTM B633	SAE AMS2451/5	SAE J514
ASTM B695	SAE AMS2451/9	SAE J1453-1
	SAE AMS2486	SAE J1453-3

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2014-032)

Review activities:
Army - AR
Navy - CG, MC, SA, SH
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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